

Living Systems® Process Suite

Frequently Asked Questions

Living Systems Process Suite Documentation

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Chapter 1

Frequently Asked Questions

1.1 General

1.1.1 What are logs?

In the context of LSPS, the term *log* can refer to one of the following:

- A log entry made when a model instances executes the Log task or `log()` function call.
Both, the task type and the function, are provided by the Standard Library and the entries are visible in the Logs view of the Management perspective and Management Console. These logs are stored in the LSPS_←_LOGS table of the LSPS system database.
- A process log entry which is automatically created by model instances.
These are logs about execution progress. They are stored in the LSPS_PROCESS_LOG table and are not visible from any client. Further information is available in the [Server Installation and Upgrade Guide](#)
- The log message of the `debugLog()` function which is displayed in standard output.

1.2 LSPS Server

1.2.1 The server returns the ConnectorTracker XA exception.

The JBoss or Wildfly server returns the following XA exception:

```
WARN [com.arjuna.ats.jta.logging.loggerI18N] [com.arjuna.ats.internal.jta.recovery.xarecovery1]
```

To remedy the situation, the database user has all required access rights:

For Oracle, make sure the Oracle user has access to the appropriate tables so they can accomplish the recovery:

```
GRANT SELECT ON sys.dba_pending_transactions TO user;
GRANT SELECT ON sys.pending_trans$ TO user;
GRANT SELECT ON sys.dba_2pc_pending TO user;
GRANT EXECUTE ON sys.dbms_xa TO user;
```

Substitute `user` with the user that connects from JBoss to Oracle. This solution works for Oracle 10g R2 (patched for bug 5945463) and 11g.

In an unpatched version older than 11g change the last GRANT EXECUTE to the following:

```
GRANT EXECUTE ON sys.dbms_system TO user;
```

1.2.2 Embedded Server starts with an Exception.

Details, such as Model Instance details, in the Management perspective which are no longer valid might be open, for example, due to a database reset, and the data is no longer present in the database. **Close any detail tab in the Management perspective.**

1.2.3 SDK Embedded Server starts with an OpenEJBException

If you get the following exception, fix the jdbc url of LSPS_DS in <YOUR_APP>-embedded/conf/conf/openejb.xml to JdbcUrl jdbc:h2:tcp://localhost/./h2/h2;MVCC=TRUE;LOCK_TIMEOUT=60000.

```
WARN JdbcServicesImpl HHH000342: Could not obtain connection to query metadata : A file path that is
    implicitly relative to the current working directory is not allowed in the database URL
    "jdbc:h2:tcp://localhost/h2/h2;MVCC=TRUE;LOCK_TIMEOUT=60000". Use an absolute path, ~/name, ./name, or the baseDir set
    [90011-197]
Exception in thread "main" org.apache.openejb.OpenEJBException: Creating application failed:
    /home/eko/testing/3.2.1038/ekassa/ekassa/ekassa-embedded/Launcher.EAR: org.apache.openejb.OpenEJBRuntimeException:
    javax.persistence.PersistenceException: [PersistenceUnit: lsps-pm] Unable to build EntityManagerFactory:
    javax.persistence.PersistenceException: [PersistenceUnit: lsps-pm] Unable to build EntityManagerFactory
    at org.apache.openejb.assembler.classic.Assembler.createApplication(Assembler.java:843)
    at org.apache.openejb.assembler.classic.Assembler.createApplication(Assembler.java:677)
    at org.apache.openejb.assembler.classic.Assembler.createApplication(Assembler.java:673)
    at com.whitestain.lsp.launcher.Launcher.launch(Launcher.java:129)
    at com.whitestain.ekassa.embedded.LSPSLauncher.main(LSPSLauncher.java:27)
Caused by: org.apache.openejb.OpenEJBException: org.apache.openejb.OpenEJBRuntimeException:
    javax.persistence.PersistenceException: [PersistenceUnit: lsps-pm] Unable to build EntityManagerFactory:
    javax.persistence.PersistenceException: [PersistenceUnit: lsps-pm] Unable to build EntityManagerFactory
    at org.apache.openejb.assembler.classic.Assembler.loadPersistenceUnits(Assembler.java:988)
    at org.apache.openejb.assembler.classic.Assembler.createApplication(Assembler.java:741)
```

1.2.4 Messages in JMS queue are stuck.

In your server logging configuration, activate the debug log level for com.whitestain.lsp.engine.CommunicationServiceBean and com.whitestain.lsp.engine.ProcessAgentImpl services and inspect the log.

1.2.5 Server returns "ISPN000299: Unable to acquire lock after 15 seconds"

Infinispan on JBoss and Wildfly returns the ISPN000299: Unable to acquire lock after 15 seconds: this occurs when an operation takes too long so the session is locked and fails to acknowledge the Vaadin heartbeat from the Application User Interface.

To remedy the problem, do the following:

- set the active-timeout parameter of the server to 10m in the profile xml file, such as, standalone-full.xml:

```
<cache-container name="web" default-cache="passivation" module="org.wildfly.clustering.web.i
    <local-cache name="passivation">
        <!-- CHANGED the acquire-timeout here: -->
        <locking isolation="REPEATABLE_READ" acquire-timeout="600000"/>
        <transaction mode="BATCH"/>
        <file-store passivation="true" purge="false"/>
    </local-cache>
</cache-container>
```

- increase the heartbeat interval in your Application User Interface in the web.xml of the application:

```
<param-name>heartbeatInterval</param-name>
<param-value>300</param-value>
```

1.3 Disabling Update of the Database Schema of a Data Type Model

To prevent update of database schema when you upload a data-type model, you have the following options:

- Disable schema update on the LSPS server altogether from **PDS** or from a **database or JMX client**.
- Disable schema update for a particular **shared record** or **relationship** on their *DB Mapping* tab.

1.4 Application User Interface

1.4.1 Browser displays an error when accessing the application.

Clean cache of your browser.

1.4.2 The session of the Application User Interface is expired.

Make sure your browser does not use out-of-date cookies and cached data: **Clean the cookies and cached data**.

1.4.3 The user remains logged in even after the session timeout period has lapsed.

Make sure your **LspsVaadinApp** servlet in the web.xml of the Application User Interface contains the `closeIdleSessions` parameter:

```
<!-- Vaadin servlet -->
<servlet>
  <servlet-name>LspsVaadinApp</servlet-name>
  ...
  <init-param>
    <param-name>closeIdleSessions</param-name>
    <param-value>true</param-value>
  </init-param>
</servlet>
```

This prevents Vaadin from sending heartbeats to the server after the session-timeout period. The session timeout is set in minutes in web.xml.

1.5 PDS

1.5.1 Icons are missing in the top toolbar of PDS on my Linux machine.

This is caused by Eclipse issue [497705](#). To remedy the situation, create a new PDS launcher:

```
#!/bin/bash
export SWT_GTK3=0
./lsps-design_64
```

1.6 Modeling

1.6.1 When should a Module be executable?

An executable Module can become a model instance, while a non-executable module is intended as a resource and cannot be instantiated: If you have a module where you define the data model only (like in the model in the MVC pattern), it is not intended for execution. Hence it does not require the execution flag.

Note that if you import an executable module to another executable module, and you instantiate the importing module (the parent), the imported executable module will be instantiated as part of the model instance.

1.6.2 There are validation problems even though the expressions are correct.

- In the GO-BPMN Explorer, right-click the module with the problem and select **Validate**.
- Clean the project: on the main menu, go to **Project -> Clean**

1.6.3 I cannot access a local variable in another local variable.

Make sure that the variable is in a context the other variable can access. If working with local variables of the same context, make sure the variable is already initialized: variables are initialized in the order they are defined in (make sure, the variables are not sorted alphabetically when using the Outline view).

1.6.4 How do I load and work with a file from my workspace.

- To work with a file, load it to the File record:

```
def File file := getResource(module -> "my-module", path -> "file.pdf");
//to access the content binary: file.content.toString("UTF-8")
```

- To work with a file *in a form*, load it into a FileResource object with the `getResource()` function:

```
new FileResource({ -> getResource("form_components", "output/myfile.txt")})
```

1.6.5 An expression returns only the last collection item, last list item or last set item.

Check if you are not using a semi-colon (;) instead of a comma (,) as a separator of the collection items: the system returns the last expression as the collection content.

Typical case of a misplaced semi-colon

```
new CategoryDataSeries([
  //semi-colon at the end of the following line causes that
  //only the next line is returned as the content
  //of the collection:
  new CategoryDataSeriesItem("Pears", 1, 5);
  new CategoryDataSeriesItem("Apples", 2, 6);
]);
```


1.6.6 I want to create a model instance over an entity.

If you want to create a model instance over an entity, pass the shared Record to the `createModelInstance()` function:

```
createModelInstance(true, thisModel(), mySharedRecord := new MyRecord() , null);
```

You can then get the Record with a `getProcessEntity(MyRecord)` function call.

1.6.7 I want to get all Todos of a model instance.

To get to-dos of a model instance, create a query with the model instance as its parameter that return the Todo type and define a condition that will exclude to-dos of other model instances, for example, `todoIterator.model← InstanceId == modelInstanceParameter.id;`

1.6.8 What is the difference between the Upload As and Run As module command?

- **Upload As** compiles and uploads the selected module and its module imports to the LSPS Module Repository of the server your PDS is connected to. In PDS, the modules uploaded to the server are managed in the **Module Management view** of the Management perspective.
- **Run As** does the same and in addition creates a model instance of the executable module. Hence this option is not available for modules that are no executable since these cannot be instantiated. In PDS, the model instance that have been executed by the server can be managed from the **Model Instances view** of the Management perspective.

1.6.9 Data Types

1.6.9.1 How do I get all shared records of a type without creating a query?

Use the `findAll()` function of the Standard Library to get all the shared records of a particular type. Also note you can use the `countAll()` function to get the number of the shared records.

1.6.9.2 Proxy Writes Directly to Database or Does not Merge.

1. Make sure the proxy object really exists:

- if you are working with a related record instance, you need to explicitly pass it as the `properties` parameter when creating the proxy instance with the `proxy()` call.
- if you are working with a record with a relationship (the record instance does not exist), you need to explicitly create a proxy for the related record and assign it to your proxy.

2. In Forms forms, make sure the proxies are created before they are used: use the `preCreateWidget()` method to create the proxies.

3. Check if your Record name does not collide with a system Record name.

1.6.9.3 Warning "Missing database index on foreign key" on a Data Relationship

Create indexes for foreign keys on the data relationship ends: select the relationship and in the Properties view, define the indexes on the Indexes tab. Alternatively, you can generate them automatically: in the Outline view, on the Tree tab, click the root node *Data Types* and in its Properties view click the **Generate Indexes** button.

How Was Database Schema of My Module Updated?

You can find information on how the schema was changed in the **Module Management** view of the Management Perspective: click the Schema Update Scripts tab.

1.6.9.4 Model upload fails with a hibernate exception.

Check whether the data type model meets the following:

- The hierarchy uses "Each record to own table".
- The top record has no property, only primary key.
- There is a record with one-to-set relation to the top record.
- The relation is one-directional.
- The relation is sorted according to the primary key of the top record.

This occurs due to an issue in Hibernate. To work around it, do one of the following:

- Add a property to the top record.
- Remove sorting on the relationship.
- Make the relationship bi-directional (the top record will have a property)

1.6.10 UI Forms

1.6.10.1 My local form variable has incorrect value.

If you are assigning a value to your variable in an `InitListener`, make sure the component with the `InitListener` is actually displayed. `InitEvents` are fired only on component display.

1.6.10.2 I cannot change properties of a listener.

Make sure the `Listener` is defined by `expression` option on the *Expression* tab is *not* selected.

1.6.11 Forms

1.6.11.1 Filter on a table column is not displayed.

If you are using the `None` value in the Filter value provider, make sure the type of the column value provide on the Detail tab of the column defines its value (in the case of `Identity`, the value should be set to the type, such as `String`).

1.6.11.2 Geolocator in the experimental forms module is missing.

- The Geolocator component has been substituted with the `Forms.detectLocation()` method.

1.6.11.3 A document UI definition reports the error "Type mismatch. Cannot convert from <> to human::UIDefinition".

- Import the *forms* module into your module. Check if your Record name does not collide with a system Record name.

1.6.11.4 Charts ignore plot options.

If a data series does not plot correctly, consider defining the values for all properties explicitly, for example, if you are changing the color of a decimal series, apart from the color parameter, set also the *range* to `true` and *lineStyle* to `lineStyle.solid`:

```
c.setPlotOptions(new PlotOptionsLineArea(color -> new Color(0, 0, 0), lineStyle -> LineStyle.solid
```

1.6.11.5 Modeling ID is ignored.

This happens on the root components of the Expression component: the modeling id of the root component returned by the expression in the Expression component uses the modeling id as set on the Expression Component in its properties. Calls to `setModelingId()` on the component that is the content of the Expression component are ignored.

1.6.12 Processes

1.6.12.1 I get "A parametric process cannot be instantiated automatically." on my process

A process that has parameters cannot run without its parameters; hence, it can be instantiated only by another entity which provides the parameters. To solve the situation, display the properties of the process and on the Detail tab unselect *Instantiate Automatically*.

1.6.12.2 I want to instantiate a process from a document.

A document is not part of a model instance and process instances can exist only as part of a model instance. Hence this is not possible. However, you can create a model instance with the `createModelInstance()` call.

1.6.13 Roles

1.6.13.1 How do I check if the current user has a role?

Typically when setting access rights for to-dos and documents, you want to check if the user who is working with the application at the given moment has the required modeled role. Use the `isPersonIn()` function.

```
//Example setting of access rights:  
isPersonIn( getCurrentPerson(), common::Client() )
```

1.6.14 RESTful Webservices

I get "The restful Webservice definition is not valid. See Problems view for details." dialog box.

Close and reopen the editor.

1.6.15 Expressions

1.6.15.1 I get "The symbol <TYPE> is undefined." validation error.

Validation reports an error of an undefined symbol for an expression that should result in a Type: This happens if the type contains <>. To solve the problem, wrap the type in the `type()` function.

```
type(Reference<String>)
```

1.6.15.2 I want to format a Date.

To format a Date object, call the `formatDate()` method with the date format as its parameter; the formatter uses the `DateTimeFormat` with the pattern and time zone.

Example call

```
now().formatDate("EEEEEEEEEEE, MMM d", "Pacific/Auckland")
```

1.6.15.3 I want to perform action on each item of a Collection or Map.

Though you can use the `foreach` construct, consider the `collect()` function of the Standard Library, especially if you create new collections in the cycle: the `collect()` function runs a closure on each item and returns a list with the outputs from the closures.

Example over List

```
//list with lists of values and names for the ListDataSeriesItem:
def List<List<Object>> itemList1 := [[1, "I am item 1 with value 1"], [3, "I am item 2 with value 1"],
~
def ListDataSeries ls1 :=
//Collect returns a list of ListDataSeriesItem; it is then passed to the ListDataSeries constructor
new ListDataSeries (collect(itemList1, { i -> def ListDataSeriesItem li := new ListDataSeriesItem(i, i)}
```

Example over Map

```
//map of values and names for the ListDataSeriesItem:
def Map<Decimal, String> mapsOfItems := [1 -> "I am item 1.", 3 -> "I am item 2.", 1.3 -> "I am item 3."]
//Creating a list of data series:
[
  new ListDataSeries (collect (mapsOfItems,
    { k,v -> def ListDataSeriesItem li := new ListDataSeriesItem(k, v)}
  )
)
]
```

1.7 Database

1.7.1 MS SQL server reports deadlocks.

If your MS SQL Server reports deadlocks, the problem might be caused by orphaned transaction: make sure to kill any hanging orphaned transactions that block database resources if such transactions are present.

Example of a select that returns orphan transactions

```
select request_owner_guid as [UoW ID] from sys.dm_tran_locks where request_session_id = -2
```

1.7.2 Migration or initialization of LSPS tables failed on MySQL.

When running on MySQL 5.6 with multibyte encoding (UTF-8), check if your database uses Barracuda as file format: `SHOW VARIABLES LIKE 'innodb_file_format'` If it is not the case, set the global `innodb_file_format=Barracuda`; set `global innodb_large_prefix=on`; `ROW_FORMAT DYNAMIC`

If you are not using Barracuda, the initialization of the LSPS database will fail with the exception:

```
SEVERE: Migration of schema 'lsp' to version <DB_VERSION> failed! Please restore backups > and roll back database and code!
```

This occurs because MySQL uses the InnoDB engine which restricts the column length and LSPS requires wider columns than allowed by MySQL.

1.8 Miscellaneous

1.8.1 How should I structure my project?

The recommended best practise is to structure your project as follows:

- java folder with customized Application User Interface resources (LSPS Application);
 - model folder with the following GO-BPMN modules:
 - main: imports all modules and does not contain any resources itself
 - common: resources required by all parts of the process, such as, data type models, general queries and functions, constraints, cache regions, etc.
 - init: module with processes that initialize the business data, that is, create your users, assign the users respective rights, update or create persisted business data, instantiate models, etc.
 - modules with your processes and other resources;
-

Example project structure

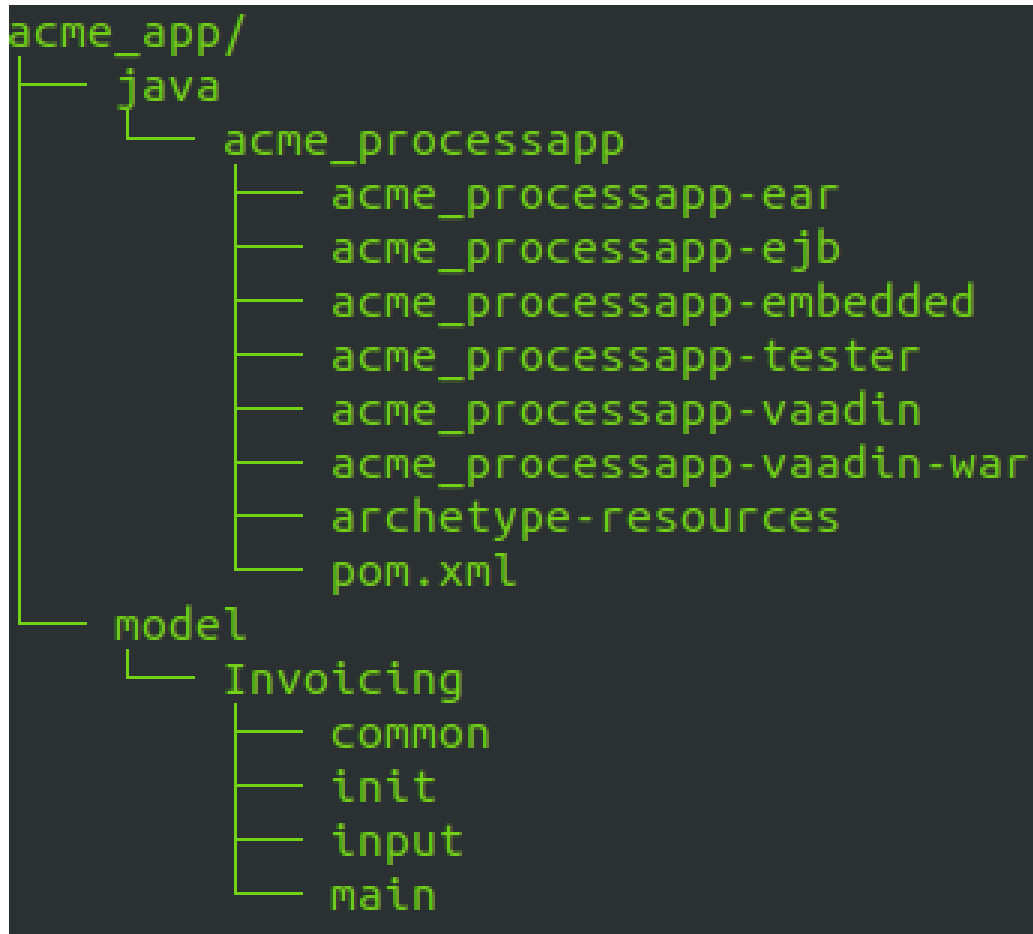


Figure 1.1 Recommended project structure

1.8.2 LSPS launcher icon is displayed incorrectly in the launcher bar in Gnome.

In the `lsps-design_64.ini` in `$PDS_HOME`, add the following **before** `vmargs`:

```
--launcher.GTK_version
```

Alternatively, add set the `SWT_GTK3` environment variable to 0:

```
export SWT_GTK3=0
```

1.8.3 To-dos and Documents is temporarily unavailable.

If a to-do or document returns an error similar to the following, **reset the to-dos** and delete the relevant documents from the database (currently no API for this action is provided).

```
Processing failed. The to-do is temporarily unavailable.
```